

Intelligent Systems,  
MIPT

# AI Agent: base practice

L11 / 07.05.26

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 [Course page](#)

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# Agenda

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## Lecture 11: Intro to AI Agent

- The concept of Agent
- Properties defining an Agent
- Training Agent
- Benchmarks & Tools

# Origins

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## **Philosophy & Cognitive Science (1950–80s)**

Norbert Wiener's cybernetics: system receives feedback from environment and adjusts behavior.  
John Searle, 1980: can a machine truly "understand" and "act"?

## **Classical AI & Multi-Agent Systems (1980–2000s)**

Minsky's "Society of Mind" (1986): intelligence as interacting agents.  
Wooldridge & Jennings (1995) formal definition

## **Reinforcement Learning (1990–2010s)**

Sutton & Barto (1998): agent = policy(state) → action, maximizes long-term reward.  
AlphaGo (2016), AlphaStar (2019) — game-playing agents

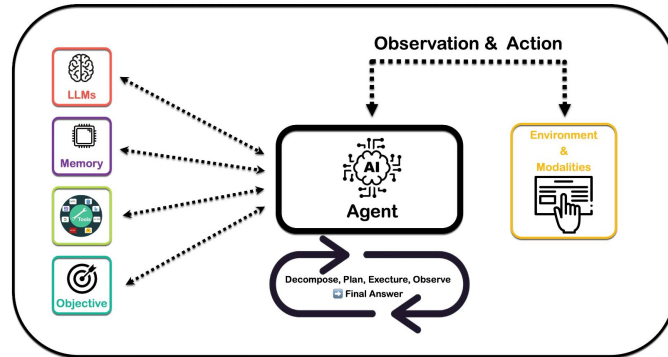
## **What changed with LLMs?**

Before 2022, an Agent was a specialized system with a hard-coded domain. ChatGPT showed that a language model can be a universal agent core

# What is an LLM Agent?

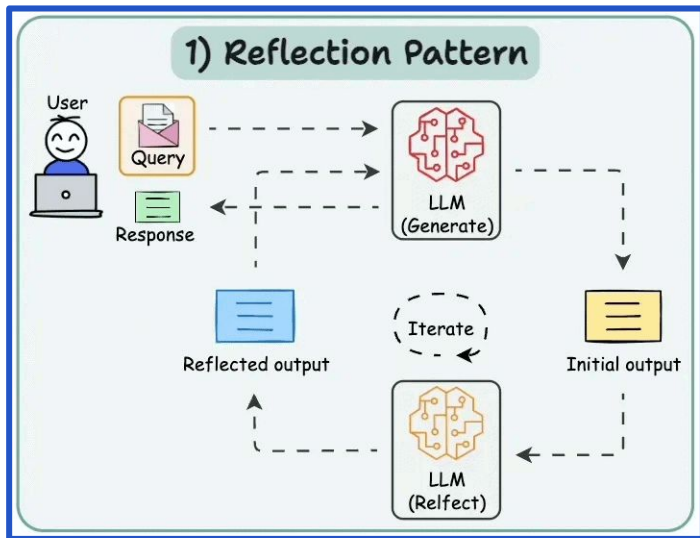
*An LLM Agent is a system where a language model acts as the brain: it perceives observations, reasons, decides on actions, and iteratively pursues a goal.*

## AI Agents / Agentic Applications

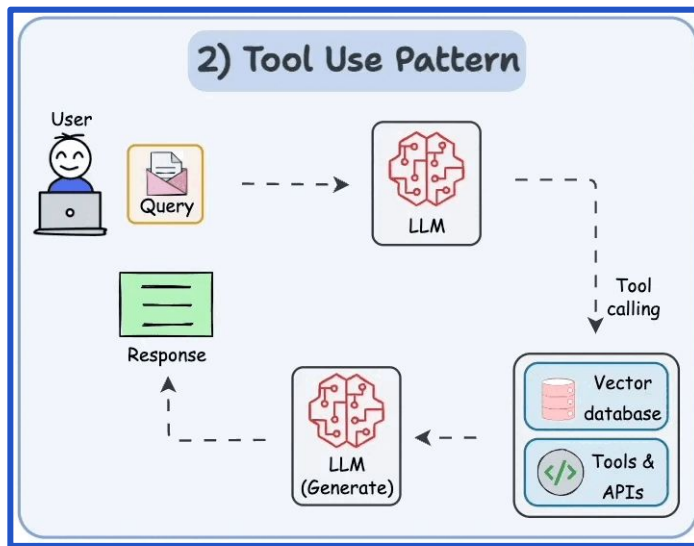
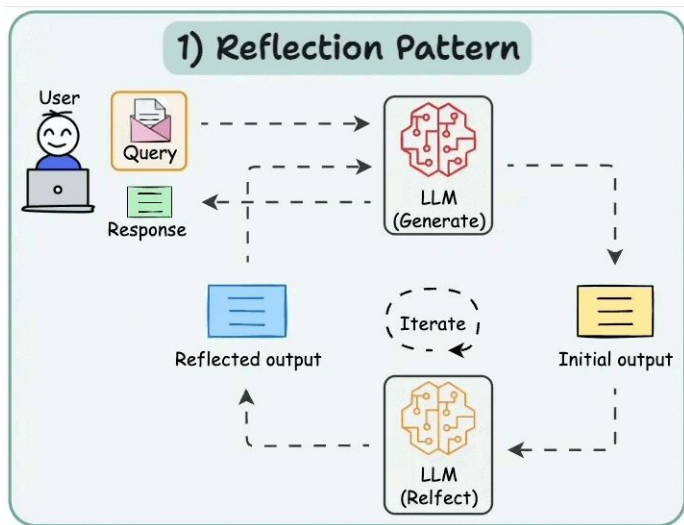


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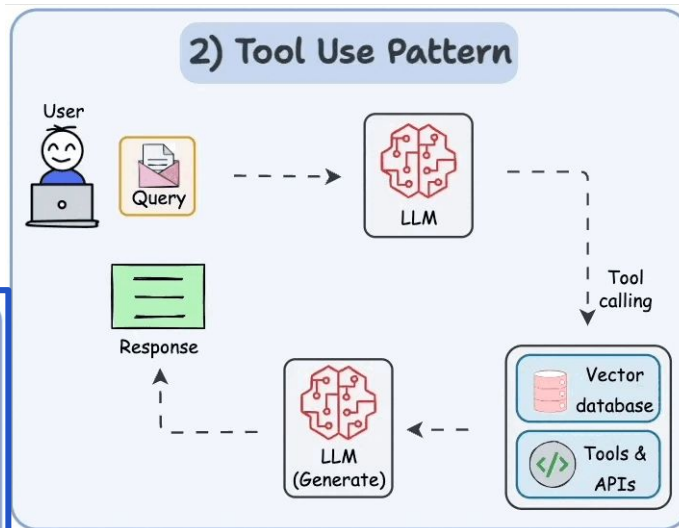
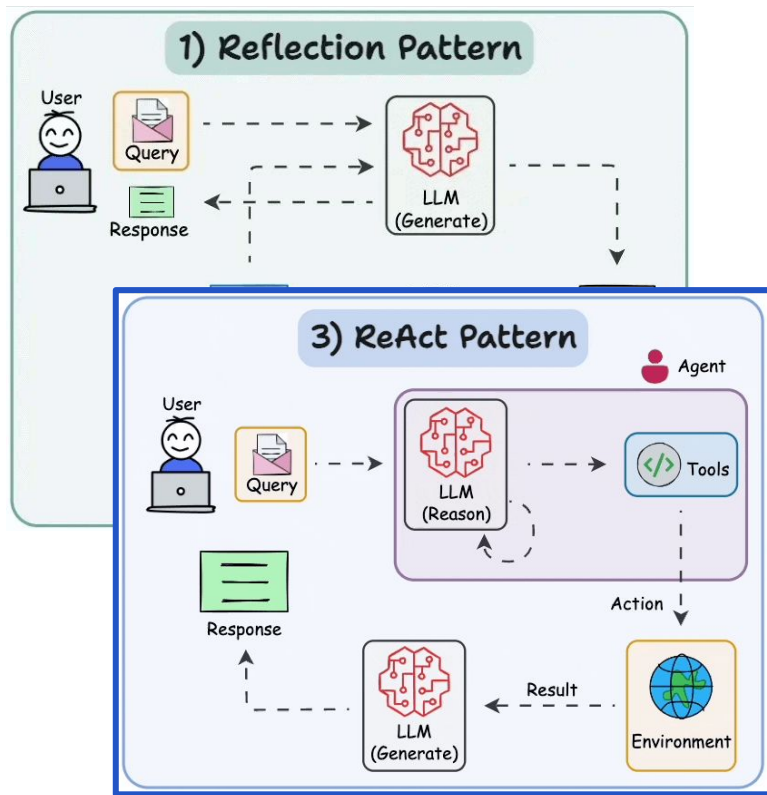
# Patterns



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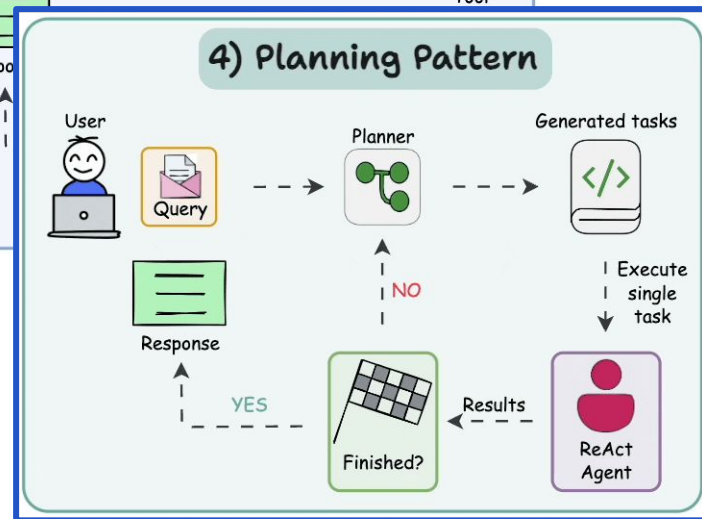
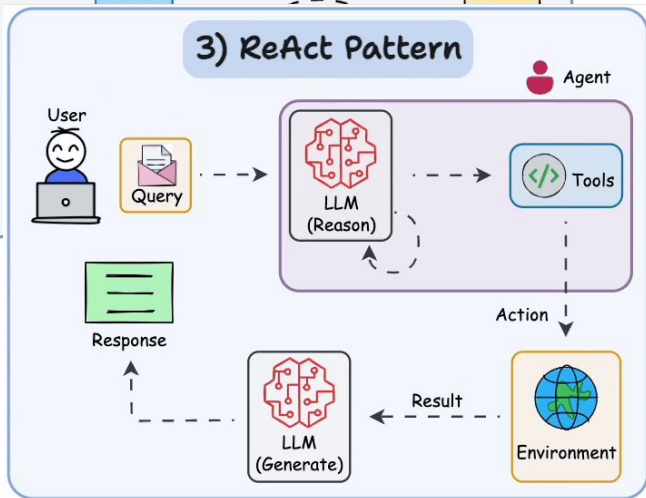
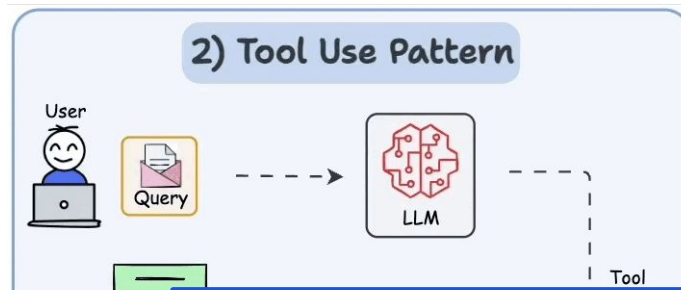
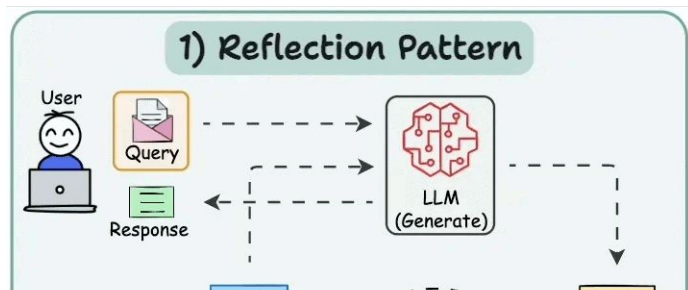


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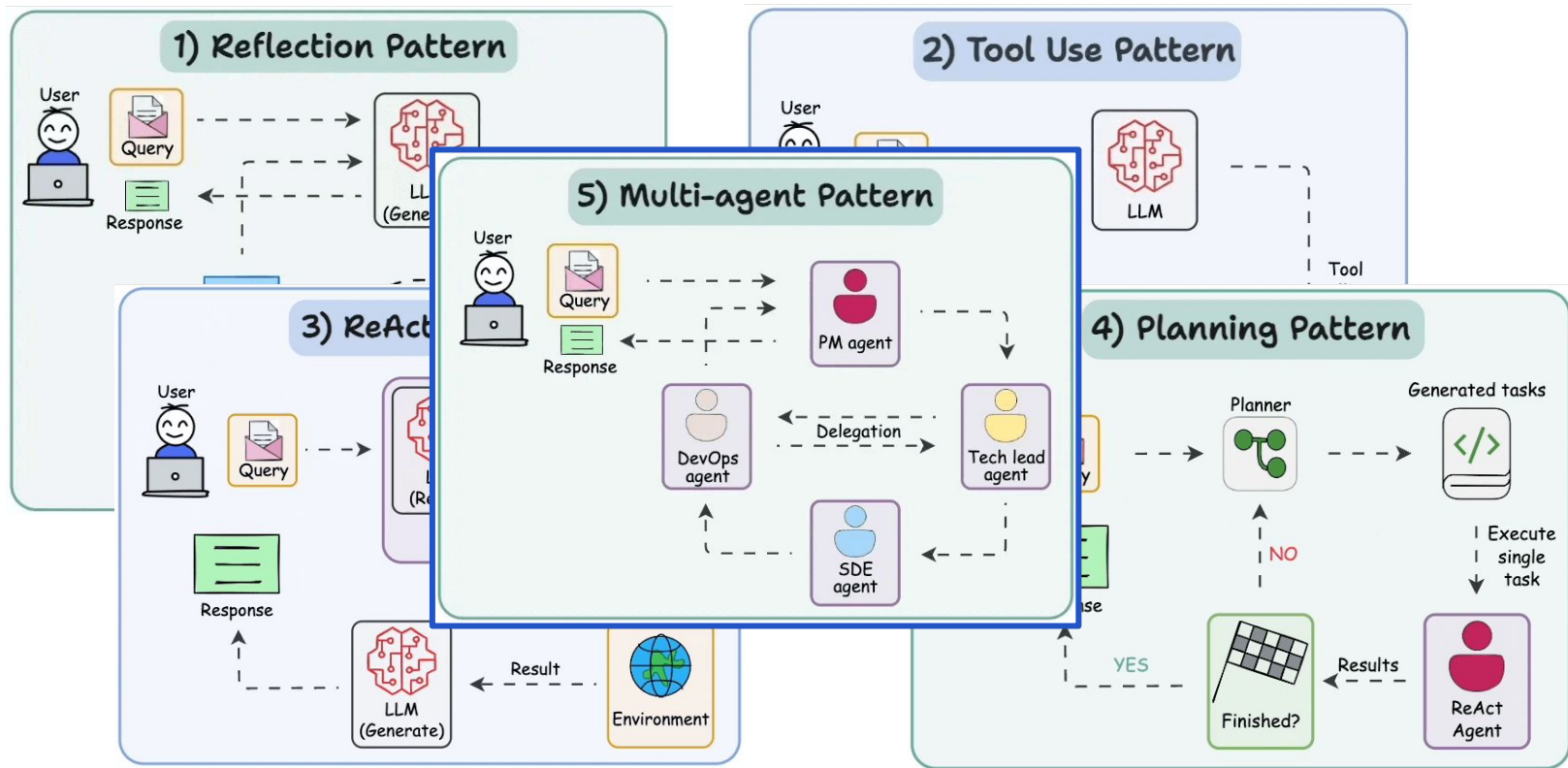
<https://blog.dailydoseofds.com/>

# Patterns



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# Type of Memory

## Short-Term Memory

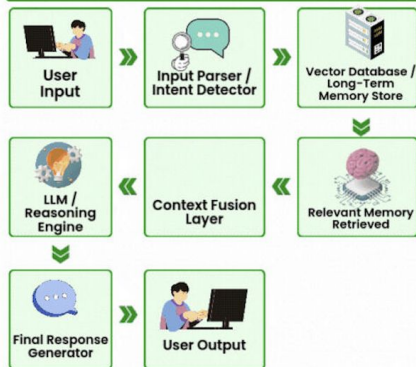
Architectural Flow



- This is the agent's working memory. It holds temporary information like recent prompts, responses, or tasks during a session. Once the interaction ends, this memory is usually cleared.
- Useful for managing live conversations and tracking steps in a task.

## Long-Term Memory

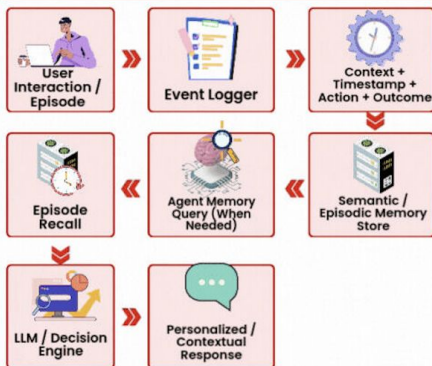
Architectural Flow



Unlike short-term memory, this retains facts and user preferences across sessions, using vector databases or knowledge graphs.

## Episodic Memory

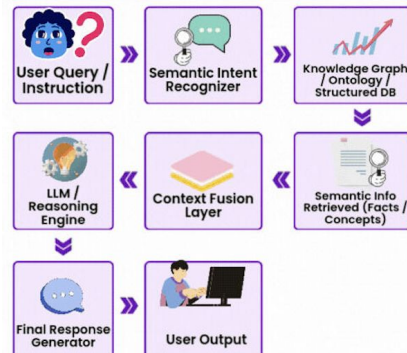
Architectural Flow



Episodic memory lets AI recall past interactions—what, when, and why—helping build context and adapt over time.

## Semantic Memory

Architectural Flow



This memory stores general facts and knowledge, enabling AI to reason, make decisions, and provide accurate answers.

# Summary

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|   |                                                                                       |                                |
|---|---------------------------------------------------------------------------------------|--------------------------------|
| 1 | <b>Accept a role</b> and perform a task within its framework                          | Role                           |
| 2 | <b>Understand</b> the request, <b>plan</b> and execute the task                       | <b>Reasoning</b><br>Reflection |
| 3 | <b>Have knowledge of the context and history</b> of who is being served by the action | Memory                         |
| 4 | <b>Have knowledge</b> in a given subject area                                         | Domain knowledge               |
| 5 | Perform actions <b>independently</b>                                                  | <b>Autonomy Action</b>         |

# How Agent trained?

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1

## Pretraining

Model sees vast internet corpus including API docs, tutorials, and code. Learns to call functions from examples — even without explicit training on agents.

2

## SFT on Agent Data

Format: {task, trajectory} where trajectory = sequence of (Thought, Action, Observation) with a correct final answer. Data: synthetic (GPT-4 traces) + human demonstrations.

3

## RLHF

Reward model learns to distinguish good vs bad trajectories. PPO or DPO optimizes the policy. Key challenge: reward in agentic tasks is often binary (solved / not solved).

4

## Tool-Specific FT

Dedicated training on calling specific tools (code, browser, calculator), parsing results correctly, and handling tool errors gracefully.

# Key papers

## ReAct Trace Example

**Task:** How many films did the director of Interstellar make?

**Thought:** Interstellar was directed by Christopher Nolan. I need his filmography.

**Action:** `search("Christopher Nolan filmography")`

**Observation:** Nolan directed 12 feature films: Memento, Dark Knight...

**Thought:** 12 films found.

**Answer:** Christopher Nolan directed 12 feature films.

## ReAct (Yao et al., NeurIPS 2022)

Interleave free-text Thought with structured Action. Grounds reasoning on real observations — fewer hallucinations. HotpotQA +6.9% vs best CoT baseline.

## Reflexion (Shinn et al., NeurIPS 2023)

After failure, generate a verbal reflection and store in episodic memory. "Learning" without gradient descent. HumanEval: 67% → 91%.

## Tool Learning (Qin et al., 2023)

Taxonomy: physical, virtual (API/code/search), and LLM tools. Key finding: tool description quality matters more than prompt length. Poor description = 30–40% drop.

# Benchmarks

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## 1. SWE-bench Verified (Software Engineering)

- Что измеряет: Способность агента решать реальные задачи в GitHub-репозиториях (фикс багов, фичи)

## 2. GAIA (General AI Assistant)

- Что измеряет: Решение сложных многошаговых задач из реального мира (анализ PDF, поиск в вебе, вычисления)

## 3. AgentBench / OSWorld

- Что измеряет: Взаимодействие с ОС и приложениями (браузер, терминал, офисные программы)

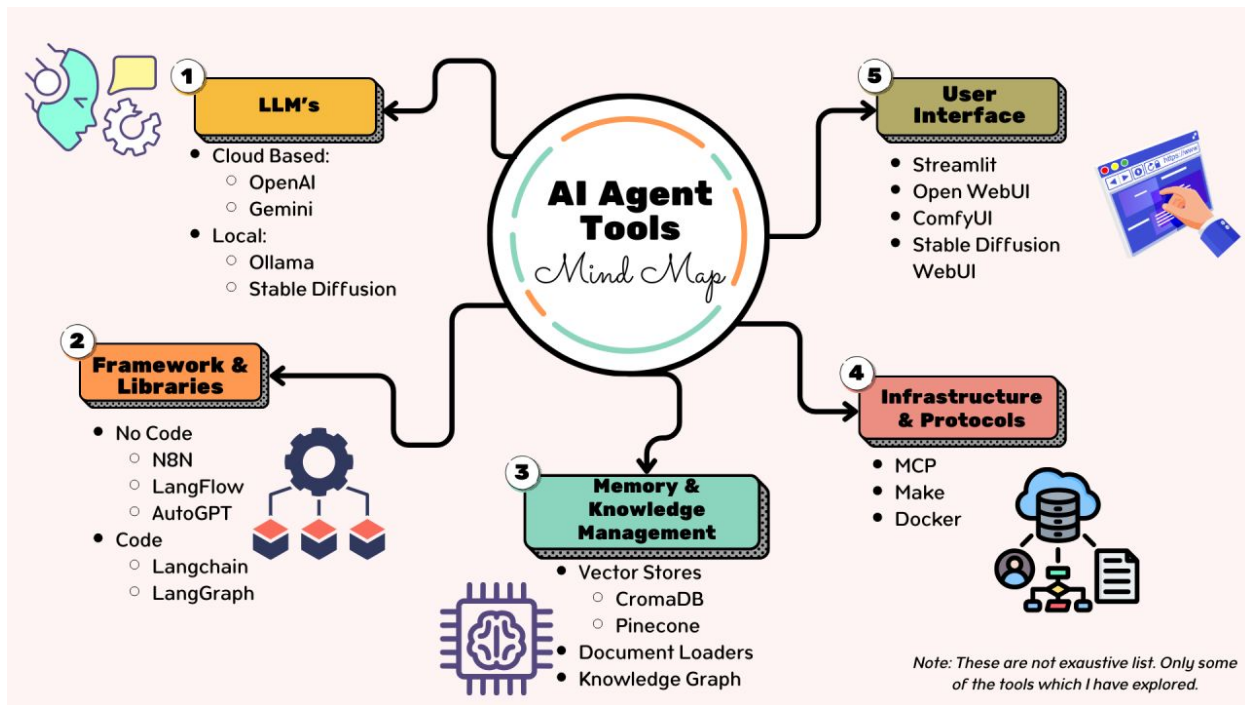
## 4. WebArena

- Что измеряет: Автономное выполнение задач в веб-среде (покупка билетов, бронирование, заполнение форм)

## 5. MLE-Bench (Machine Learning Engineering)

- Что измеряет: Способность агента участвовать в соревнованиях по ML (Kaggle-style): от анализа данных до обучения моделей

# Frameworks



# Go Testing



Ссылка на ноутбук:  
[Ссылка тут](#)



# Conclusion

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- **Concept origins**
- **Core Architecture**
- **Training Pipeline**
- **Benchmarks & Tools**
- **Testing frameworks**